CSS 2025: Large-Scale IoT Device Identification via HTTP Basic Auth Responses
HTTP Basic Authentication in Global IoT & Network Infrastructure
Billions of connected devices respond to probes with HTTP/1.1 401 Unauthorized;
The response header contains a tiny string:
WWW-Authenticate: Basic realm="[String]"
The Realm String Ambiguity
• ”Broadband Router” or ”IPC-HFW”?
• Cryptic abbreviations and vendor jargon;
• Handcrafted regular expressions fail on tens of thousands of obscure brands
LLM-Powered Semantic Inference
• Leverages world knowledge of hardware specs;
• Disentangles brand acronyms and code names;
• Extracts vendor, device category, & model
YNU SPIRAL Innovation (CSS 2025: Otsuka, Yin Minn Pa Pa, Yoshioka)
Massive IoT Device Identification via Basic Auth Response Analysis
Applies LLM semantic parsers to Internet-wide scan datasets (Censys / Shodan);
Achieves high-accuracy classification across previously unidentifiable millions of edge nodes
Scalable Threat Intelligence & National Vulnerability Triage
Converts a single lightweight HTTP banner header into actionable hardware intelligence;
Allows rapid assessment of newly announced CVE impact across millions of connected devices